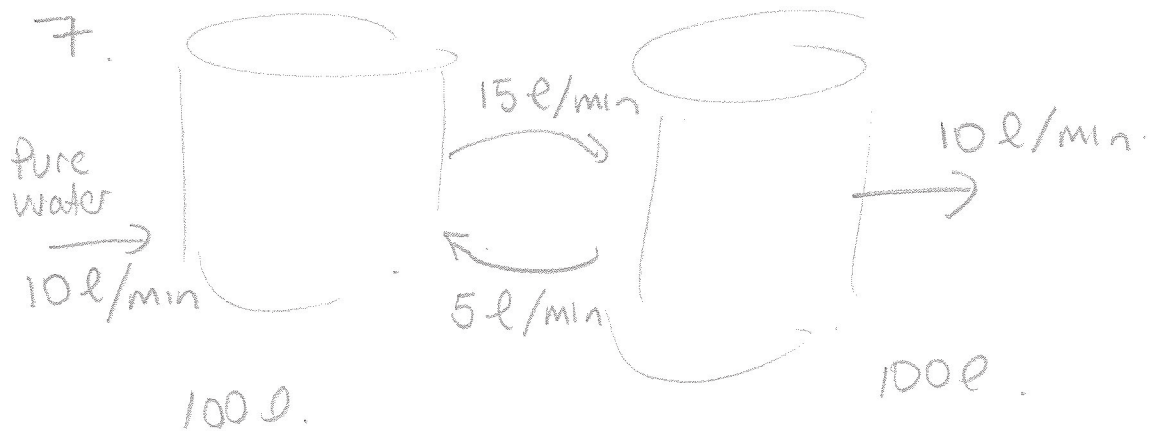


Tutorial 5.

7.



Tank 1

Tank 2

initially; 50 kg salt.

$y_i(t)$ is conc
of salt in Tank i , $i=1, 2$.

$$y_1'(t) = \frac{5}{100} y_2(t) - \frac{15}{100} y_1(t)$$

$$\begin{aligned} y_2'(t) &= \frac{15}{100} y_1(t) - \frac{5}{100} y_2(t) - \frac{10}{100} y_2(t) \\ &= \frac{15}{100} y_1(t) - \frac{15}{100} y_2(t) \end{aligned}$$

$$\begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix} = \begin{pmatrix} -0.15 y_1 + 0.05 y_2 \\ 0.15 y_1 - 0.15 y_2 \end{pmatrix} = \begin{pmatrix} -0.15 & 0.05 \\ 0.15 & -0.15 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Find λ ;

$$\det \begin{pmatrix} -0.15 - \lambda & 0.05 \\ 0.15 & -0.15 - \lambda \end{pmatrix} = 0$$

$$\begin{aligned} (\lambda + 0.15)^2 - 0.05(0.15) &= 0 \\ \lambda^2 + 0.3\lambda + 0.15^2 - 0.05(0.15) &= 0 \end{aligned}$$

Total 5

$$7. \text{ ctd. } \lambda^2 + 0.3\lambda + (0.15)(0.15 - 0.05) = 0$$

$$\lambda^2 + 0.3\lambda + (0.15)(0.1) = 0$$

$$\lambda = \frac{-0.3 \pm \sqrt{(0.3)^2 - 4(0.15)(0.1)}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.03}}{2} = \frac{-0.3 \pm 0.173}{2}$$

$$\lambda = -0.237 \text{ or } -0.0634$$

$$\text{so if } \lambda = -0.237$$

$$-0.15y_1 + 0.05y_2 = -0.237y_1$$

$$0.05y_2 = -0.087y_1$$

$$y_2 = -1.74y_1$$

$$\begin{pmatrix} 1 \\ -1.74 \end{pmatrix}$$

$$\text{if } \lambda = -0.0634$$

$$-0.15y_1 + 0.05y_2 = -0.0634y_1$$

$$0.05y_2 = 0.0866y_1 \quad \left(\frac{0.173}{2}\right) \quad \begin{pmatrix} 1 \\ 1.74 \end{pmatrix}$$

$$y_2 = 1.732y_1$$

$$y(t) = c_1 e^{-0.0634t} \begin{pmatrix} 1 \\ 1.74 \end{pmatrix} + c_2 e^{-0.237t} \begin{pmatrix} 1 \\ -1.74 \end{pmatrix}$$

$$\text{at } t=0 \quad C_1 + C_2 = 0 \quad C_2 = -C_1$$

$$1.74 (C_1 - C_2) = 50$$

$$1.74 \cdot 2C_1 = 50$$

$$C_1 = \frac{25}{1.74}$$

$$1.74^2 = 3.0276$$

1.74 must be
approx for $\sqrt{3}$

$$\text{so } y_1(t) = \frac{25}{\sqrt{3}} e^{-0.0634t} - \frac{25}{\sqrt{3}} e^{-0.237t}$$

$$\lambda = \frac{-0.3 \pm \sqrt{(0.3)^2 - 4(0.15)(0.1)}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.09 - 0.06}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.03}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.01} \sqrt{3}}{2} = \frac{-0.3 \pm 0.1\sqrt{3}}{2}$$

$$= -0.15 \pm 0.05\sqrt{3}$$

Tutorial 5:

$$\lambda = -0.15 + 0.05\sqrt{3}$$

find

$$-0.15y_1 + 0.05y_2 = -0.15y_1 + 0.05\sqrt{3}y_1$$

$$y_2 = \sqrt{3}y_1 \quad \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$\lambda = -0.15 - 0.05\sqrt{3}$$

$$\text{Then } -0.15y_1 + 0.05y_2 = -0.15y_1 - 0.05\sqrt{3}y_1$$

$$\text{so } y_2 = -\sqrt{3}y_1 \quad \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}$$

$$\text{so } y(t) = c_1 e^{-0.0634t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + c_2 e^{-0.236t} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$\text{and } y(0) = \begin{pmatrix} 0 \\ 50 \end{pmatrix} \quad \text{so } c_1 + c_2 = 0$$

$$\text{and } 50 = -\sqrt{3}c_1 + \sqrt{3}(-c_1)$$

$$\text{so } \frac{50}{-2\sqrt{3}} = c_1 = -\frac{25}{\sqrt{3}}$$

$$\text{and } c_2 = \frac{25}{\sqrt{3}}$$

$$y(t) = -\frac{25}{\sqrt{3}} e^{-0.0634t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + \frac{25}{\sqrt{3}} e^{-0.236t} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$y_1(t) = -\frac{25}{\sqrt{3}} e^{-0.0634t} + \frac{25}{\sqrt{3}} e^{-0.236t}$$